Dyck paths with infinite support and Unbounded Temporal Logic

Introduction

This work extends Ferrari's [1] correspondence between Dyck lattices, Heyting algebras, and interval temporal logic from finite to infinite discrete time. Ferrari sketched two candidate notions of infinite Dyck path; I make both precise — one as the completion of the family of finite Dyck lattices, the other as the space of paths with infinite support — and compare their algebraic behavior. The main observation is an asymmetry: the first construction yields a complete Heyting algebra outright, while the second loses the bottom element and closure under meets. I study the resulting structures and the temporal logics they support, and formulate the associated open problems.

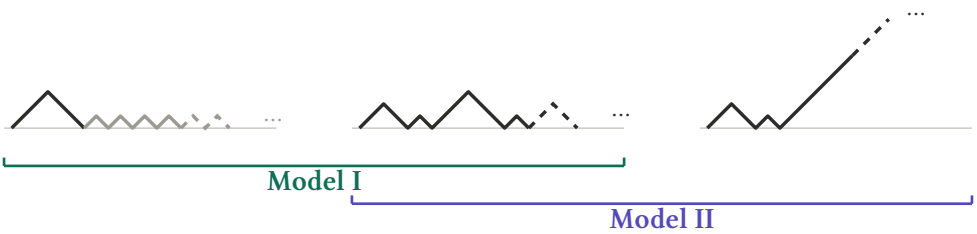


Figure 1: Three kinds of infinite path: finite content followed by flat peaks forever; excursions recurring without end; finitely many rebounds, then unbounded ascent.

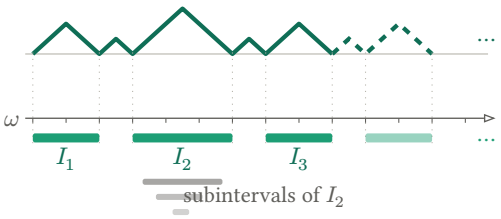


Figure 2: Each excursion of the path occupies a stretch of time; the stretches form an antichain $I_1, I_2, I_3, \ldots$ of intervals of ω, and the element of the algebra is its down-closure. Flat peaks occupy no stretch.

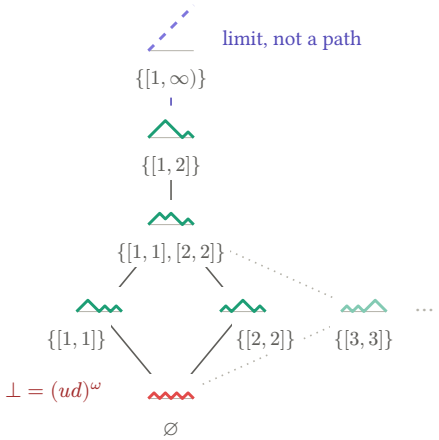


Figure 3: A fragment of the infinite Dyck lattice, in the style of Ferrari's Figure 3. Each node carries a path and, beneath it, the antichain of time intervals it determines. The two atoms occupy the instants 1 and 2 separately; their join keeps them separate, and only one step higher do they merge into the single interval $[1, 2]$. Upward the atoms never run out, and the ascending chain of pyramids converges to a ray that no path realises.

Dyck paths with infinite support

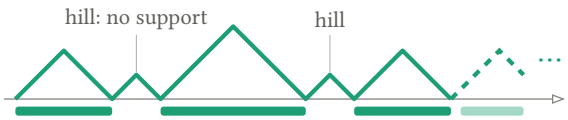


Figure 4: **Generic admissible path — support under the excursions**
bands mark positions of height ≥ 2 excursions; hills contribute nothing; supp infinite ✓



Figure 5: **The infinite hill $(ud)^\omega$ — never leaves the floor**
geometrically busy, logically empty: the path **is** the floor; supp $= \varnothing$, this element is $\bot$ ✗



Figure 6: **Join of all atoms $uu(du)^\omega$ — punctual truth everywhere**
one step above the floor: height 2 cleared at every position; supp $= \omega$, admissible ✓



Figure 7: **D (teal): excursions over even positions — E (amber): over odd positions**
both supports infinite ✓✓ — but disjoint: the bands interleave and never overlap



Figure 8: **$D \wedge E$ = pointwise minimum**
the min is on the floor wherever **either** path is — with interleaved supports, that is everywhere: $D \cap E = (ud)^\omega = \bot$ ✗ — the meet of two admissible paths lands exactly on the banned element

The two models, informally

Ferrari's theory relates three things: Dyck paths of a fixed finite length, the Heyting algebras they form under the natural order, and a logic whose worlds are the intervals of a finite line of time. He observes that one may try to replay this over an infinite discrete line of time, and proposes two candidate notions of infinite Dyck path. We call the resulting constructions Model I and Model II.

Model I: paths that always come back.

One starts with nothing infinite at all, only the collection of *all* finite Dyck paths, ordered so that a path sits below another when it stays weakly underneath it. Passing from one length to the next means padding a path with a small flat peak on the right, and such padding adds no information, so paths differing only by padding are treated as the same object. This ordered collection is then completed, in the sense that every increasing family of paths is given a limit. The completion introduces genuinely infinite objects, and they turn out to have a uniform shape: they are paths that rise and fall forever, returning to the ground line infinitely often, each individual rise and fall being of finite extent. Alongside these there appear a few boundary objects that no path realises directly — the limits of paths that climb higher and higher — which the completion is obliged to supply and which behave as ideal points rather than as paths.

Model II: paths that stay busy forever.

The second construction takes infinite paths as the primitive notion instead of manufacturing them. An infinite path here is simply a path that never ends. Two decisions distinguish it from the first

construction. First, the line of time is enlarged so that unbounded stretches of time — everything from some instant onward — count as legitimate worlds at which a statement may be evaluated; this is what makes it possible to say "from this moment on, forever". Second, one refuses those paths that do something interesting only for a while and then subside into an endless run of small flat peaks. Such a path is an impostor: the flat peaks carry no content, so the path holds exactly the information of the finite path preceding them, dressed up as an infinite object. Ruling them out is what "infinite support" means: the interesting behaviour must keep recurring, no matter how far to the right one looks.

Where they differ. The first construction excludes nothing and adds ideal points; the second excludes a whole family of paths and adds real ones. Adding is harmless, removing is not. The path with no content at all — flat peaks throughout — is the least element of the first model, and its removal is precisely what the second model demands. A structure deprived of its least element cannot combine statements conjunctively without risk: two paths may each remain busy forever and yet be busy in places that never coincide, so that everything they hold in common evaporates. This is the central asymmetry between the two.

Algebraic interpretation

Write $\mathrm{Int}_b(\omega)$ for the bounded intervals of ω ordered by inclusion, $\mathrm{Int}_\infty(\omega)$ for the same enlarged by the rays $[a,\infty)$, and for a down-set D put

$$\mathrm{supp}(D) = \{x \in \omega : [x,x] \in D\}.$$

Model I

The algebra is the full down-set algebra

$$\mathcal{H}^{\mathrm{reb}} = \mathcal{O}(\mathrm{Int}_b(\omega)),$$

with $\wedge = \cap$, $\vee = \cup$ at arbitrary arity, $\bot = \varnothing$, $\top = \mathrm{Int}_b(\omega)$, and

$$D \to E = \{I : \forall J \subseteq I,\, J \in D \Rightarrow J \in E\}, \quad \neg D = D \to \varnothing.$$

This is a complete Heyting algebra. Its compact elements are the finite Dyck paths; the remaining elements of finite-interval type are the infinitely rebounding paths; the ray-type elements are non-compact suprema realised by no path. Coinductively, the paths are the inhabitants of $\nu X.(D_{\mathrm{fin}} \times X)$ where $D_{\mathrm{fin}} = \mu Y.(1 + Y \times Y)$ generates the finite excursions.

Model II

The carrier is the sub-collection of the enlarged down-set algebra consisting of the elements of unbounded support:

$$\mathcal{H}^{\mathrm{sup}} = \{D \in \mathcal{O}(\mathrm{Int}_\infty(\omega)) : \mathrm{supp}(D) \text{ unbounded in } \omega\}.$$

It is closed under arbitrary nonempty joins, and under relative pseudocomplementation, since

$$\mathrm{supp}(D \to E) = (\omega \setminus \mathrm{supp}(D)) \cup \mathrm{supp}(E) \supseteq \mathrm{supp}(E).$$

It has no least element and no minimal elements, and meets are only partial: the infimum of D and E exists exactly when $\mathrm{supp}(D) \cap \mathrm{supp}(E)$ is unbounded, and then coincides with $D \cap E$. Pseudocomplementation is unavailable, requiring a bottom that has been removed. Coinductively the carrier is a proper subset of $\nu X.(D_{\mathrm{fin}} \times X + X)$, cut out by the condition that nontrivial content be emitted cofinally often.

The logic of unbounded intervals

Both models interpret a propositional language whose worlds are stretches of time and whose implication quantifies over sub-stretches. Everything they share — the line of time, the valuation schema, the connectives that use the interval order — is fixed first, so that each model can afterwards be given by its own two decisions: which stretches of time count as worlds, and which propositions are admitted.

Common ground

Time states and intervals

Let $T = \omega = \{t_0, t_1, t_2, \ldots\}$ be an infinite discrete linearly ordered set, $t_0 < t_1 < t_2 < \cdots$, whose elements are called *time states*. Write

$$\big[t_i, t_j\big] = \big\{t \mid t_i \le t \le t_j\big\}, \quad i \le j < \omega$$

for a *bounded* interval and $[t_i, \infty) = \{t \mid t \ge t_i\}$ for a *ray*, and put

$$\mathrm{Int}_b(T) = \big\{\big[t_i, t_j\big] : i \le j < \omega\big\}, \quad \mathrm{Int}_\infty(T) = \mathrm{Int}_b(T) \cup \{[t_i, \infty) : i \in \omega\},$$

both partially ordered by inclusion. This is the first decision: Model I evaluates over $\mathrm{Int}_b(T)$ alone, so that rays are *not* intervals of T there; Model II evaluates over $\mathrm{Int}_\infty(T)$.

Valuation

Fix a set W of worlds — $\mathrm{Int}_b(T)$ or $\mathrm{Int}_\infty(T)$ — ordered by inclusion, and a propositional language $\mathcal{L}$. A valuation is a map

$$v : \mathcal{L} \longrightarrow 2^W : \varphi \mid\longrightarrow v_\varphi : W \longrightarrow \{0, 1\},$$

and we write $\varphi(I)$ for $v_\varphi(I)$. A proposition φ is *valid* when $\varphi(I) = 1$ for every $I \in W$. On whichever of the connectives $\mathcal{L}$ contains, the clauses are pointwise:

- $(\varphi \vee \psi)(I) = 1$ whenever $\varphi(I) = 1$ or $\psi(I) = 1$;
- $\big(\bigvee_k \varphi_k\big)(I) = 1$ whenever $\varphi_k(I) = 1$ for some k;
- $(\varphi \wedge \psi)(I) = 1$ whenever $\varphi(I) = 1$ and $\psi(I) = 1$;
- $(\neg\varphi)(I) = 1$ whenever $\varphi(I) = 0$;
- $(\varphi \to \psi)(I) = 1$ whenever the following holds: if $\varphi(I) = 1$, then $\psi(I) = 1$.

Infinitary disjunction is present in both models, and it is not a convenience: over an infinite time domain an antichain of intervals may be infinite, so finite disjunction no longer suffices for the normal forms below.

The order-sensitive connectives

So far the partial order of W plays no role. The connectives that use it are

- $(\Box\varphi)(I) = 1$ when $\varphi(J) = 1$ for all $J \subseteq I$;
- $(\Diamond\,\varphi)(I) = 1$ when $\varphi(J) = 1$ for some $J \subseteq I$;
- $(\varphi \rightsquigarrow \psi)(I) = 1$ when, for all $J \subseteq I$, if $\varphi(J) = 1$ then $\psi(J) = 1$;
- $(\sim \varphi)(I) = 1$ when $\varphi(J) = 0$ for all $J \subseteq I$.

In every clause J ranges over the worlds below I, so the same four definitions read differently in the two models: over $\mathrm{Int}_\infty(T)$ the quantifier reaches sub-rays as well as bounded subintervals. Note that $\sim\varphi$ is $\varphi \rightsquigarrow \bot$ and so presupposes a falsum; it is available in Model I only.

The hereditary fragment
Put

$$\Theta(W) = \{\varphi \mid \varphi \to \Box\varphi \text{ is valid}\},$$

so that $\varphi \in \Theta(W)$ exactly when truth of φ at a world carries to every world below it. As in the finite case, $\Theta(W)$ is closed under $\wedge$ and $\vee$ — and, by the same argument, under arbitrary $\bigvee$ — but not under $\neg$ nor under $\to$; witnesses are recorded in Model I, whose punctual atoms make them easy to write down. Replacing $\neg$ and $\to$ by $\sim$ and $\rightsquigarrow$ is what repairs this. Both models work inside a hereditary fragment, and which one is the second decision.

Model I — infinite rebound

Worlds are the bounded stretches of time, $W = \mathrm{Int}_b(T)$, and no proposition is excluded. Since no world witnesses the unbounded future, no "henceforth" operator is available here.

Propositions
The set ITL_ω is defined recursively:

- $\bot, \top \in \mathrm{ITL}_\omega$; and $\varepsilon_i \in \mathrm{ITL}_\omega$ for every $i \in \omega$ (the propositional variables);
- if $\varphi, \psi \in \mathrm{ITL}_\omega$, then $\varphi \vee \psi$, $\varphi \wedge \psi$, $\varphi \to \psi$, $\neg\varphi \in \mathrm{ITL}_\omega$;
- if $\{\varphi_k\}_{k\in K} \subseteq \mathrm{ITL}_\omega$, then $\bigvee_{k\in K} \varphi_k \in \mathrm{ITL}_\omega$.

In grammar form,

$$\varphi ::= \varepsilon_i \mid \bot \mid \top \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \bigvee_k \varphi_k \mid \varphi \to \varphi, \quad \text{with} \quad \neg\varphi := \varphi \to \bot$$

— the full intuitionistic signature, the infinitary join apart. The atoms are *punctual*,

$$\varepsilon_i(I) = 1 \iff I = [t_i, t_i] = \{t_i\},$$

and the remaining clauses are those fixed above, read over $\mathrm{Int}_b(T)$.

The hereditary fragment
Put

$$\Theta_\omega = \Theta(\mathrm{Int}_b(T)) \cap \mathrm{ITL}_\omega = \{\varphi \in \mathrm{ITL}_\omega \mid \varphi \to \Box\varphi \text{ is valid}\},$$

so that $\varphi \in \Theta_\omega$ exactly when truth of φ on an interval carries to all of its subintervals. The failures of closure announced above are witnessed here: $\varphi = \varepsilon_i \vee \varepsilon_{i+1}$ is hereditary while $\neg\varphi$ is not, and $(\varphi, \psi) = (\varepsilon_1 \vee \varepsilon_2, \varepsilon_2)$ is a hereditary pair whose implication is not. Both counterexamples are local and transfer verbatim from the finite case, and both are repaired by passing to $\sim$ and $\rightsquigarrow$. Writing $\overline{\varphi} = {\sim\sim}\,\varphi$ and $\varepsilon_I = \bigvee_{i\in I} \varepsilon_i$, one has as before $\overline{\varepsilon_I}(J) = 1$ if and only if $J \subseteq I$.

Quotient and normal form
The quotient $[\Theta_\omega] = \Theta_\omega/\!\Leftrightarrow$, equipped with $\wedge, \vee, \sim, \rightsquigarrow$, is a **complete** Heyting algebra, isomorphic to $\mathcal{O}(\mathrm{Int}_b(\omega))$. Ferrari's Lemma 3.7 becomes: for every $\varphi \in \Theta_\omega$ there is a unique antichain $\{I_\alpha\}$ of *generalized* intervals — bounded intervals together with at most one ray $[t_i, \infty)$ — such that

$$\varphi \Leftrightarrow \bigvee_\alpha \overline{\varepsilon_{I_\alpha}}.$$

The ray case is unavoidable and is the one genuine break with the finite statement: when the intervals on which φ holds form an ascending chain with no maximal member, no bounded interval

represents the join, and $\varepsilon_{[t_i,\infty)} = \bigvee_{j \geq i} \varepsilon_j$ is needed. Rays occur here as *disjunctions*, never as evaluation points.

Model II — infinite support

Worlds now include the unbounded stretches, $W = \mathrm{Int}_\infty(T)$, so propositions are evaluated at rays as well as at bounded intervals and "henceforth" becomes meaningful. Against that, only the propositions of unbounded support are admitted, and the fragment so cut out is not closed under all the connectives.

Propositions

We take as primitive the *hereditary interval atoms* π_I, one for each $I \in \mathrm{Int}_\infty(T)$, and close under disjunction and $\rightsquigarrow$:

$$\varphi ::= \pi_I \mid \top \mid \varphi \vee \varphi \mid \bigvee_k \varphi_k \mid \varphi \rightsquigarrow \varphi.$$

Neither $\bot$ nor $\wedge$ occurs, for the reason recorded under *Closure* below; the trade is exact — one gains a temporal modality and loses a constant together with a connective. Taking π_I primitive, rather than defining it as $\sim\!\sim\, \varepsilon_I$, is deliberate: $\sim$ requires $\bot$, which is absent here. The atoms are not punctual but hereditary by construction,

$$\pi_I(J) = 1 \iff J \subseteq I,$$

and here J ranges over sub-rays of I as well as over bounded subintervals.

Henceforth

Set $G_i := \pi_{[t_i,\infty)}$, read *we are at or after instant i*, so that

$$G_i(I) = 1 \iff I \subseteq [t_i, \infty) \iff \min I \geq t_i.$$

It is hereditary, hence lies in Θ, and it is the modality that the enlarged interval structure makes meaningful: in Model I no evaluation point is a ray, so the corresponding proposition exists but is witnessed by no world.

The admissible fragment

For $\varphi \in \Theta(\mathrm{Int}_\infty(T))$ put

$$\mathrm{supp}(\varphi) = \{i \in \omega : \varphi([t_i, t_i]) = 1\}, \quad \Theta^{\mathrm{sup}} = \{\varphi \in \Theta \mid \mathrm{supp}(\varphi) \text{ unbounded in } \omega\}.$$

Every $\varphi \in \Theta^{\mathrm{sup}}$ is equivalent to $\bigvee_\alpha \pi_{I_\alpha}$ for a unique antichain of generalized intervals with $\bigcup_\alpha I_\alpha$ unbounded — the propositional form of *the path has infinite support*, excluding exactly the paths $v(ud)^\omega$.

Closure

Θ^{sup} is closed under $\vee$ and under arbitrary $\bigvee$, since $\mathrm{supp}(\varphi \vee \psi) \supseteq \mathrm{supp}(\varphi)$; and under $\rightsquigarrow$ whenever the consequent is admissible, since

$$\mathrm{supp}(\varphi \rightsquigarrow \psi) = (\omega \setminus \mathrm{supp}(\varphi)) \cup \mathrm{supp}(\psi).$$

It is **not** closed under $\wedge$: taking φ with support the even instants and ψ with support the odd ones, both are admissible while $\mathrm{supp}(\varphi \wedge \psi) = \varnothing$. Moreover $\bot \notin \Theta^{\mathrm{sup}}$, so $\sim$ is undefined. This is why the signature above omits both.

The operators at a glance

Syntax	Semantics — $\varphi(I) = 1$ when	I	II
ε_i	$I = [t_i, t_i]$ — *punctual atom*	✓	—
π_I	$J \subseteq I$ — *hereditary atom, one per $I \in W$*	—	✓
$\bot$	never	✓	—
$\top$	always	✓	✓
$\varphi \wedge \psi$	$\varphi(I) = 1$ and $\psi(I) = 1$	✓	—
$\varphi \vee \psi$	$\varphi(I) = 1$ or $\psi(I) = 1$	✓	✓
$\bigvee_k \varphi_k$	$\varphi_k(I) = 1$ for some k	✓	✓
$\neg\varphi$	$\varphi(I) = 0$ — *i.e. $\varphi \to \bot$*	✓	—
$\varphi \to \psi$	if $\varphi(I) = 1$ then $\psi(I) = 1$	✓	—
$\sim \varphi$	$\varphi(J) = 0$ for all $J \subseteq I$ — *i.e. $\varphi \nrightarrow \bot$*	✓	—
$\varphi \nrightarrow \psi$	for all $J \subseteq I$: if $\varphi(J) = 1$ then $\psi(J) = 1$	✓	✓
$\Box\varphi$	$\varphi(J) = 1$ for all $J \subseteq I$	✓	✓
$\Diamond\varphi$	$\varphi(J) = 1$ for some $J \subseteq I$	✓	✓
G_i	$I \subseteq [t_i, \infty)$, i.e. $\min I \geq t_i$ — *abbreviates $\pi_{[t_i,\infty)}$*	—	✓

I and J range over the worlds: $W = \mathrm{Int}_b(T)$ in Model I, so J runs over bounded subintervals only; $W = \mathrm{Int}_\infty(T)$ in Model II, so J runs over sub-rays as well. The clauses themselves are unchanged — only the quantifier domain differs.

Three departures from the finite presentation, in order of severity: infinitary disjunction becomes necessary in both models; the normal form requires rays as generalized intervals in both; and only in Model II does the carrier fail to be closed under the connectives, which forces a reduced signature rather than merely a longer list of clauses.

Bibliography

[1] L. S. Ferrari, "Dyck Algebras, Interval Temporal Logic, and Posets of Intervals," *ArXiv*, vol. abs/1503.04991, 2015, [Online]. Available: https://api.semanticscholar.org/CorpusID:14279208

Appendix: Model I and Model II at a glance

	Model I — infinite rebound	**Model II — infinite support**
Description	Completion of the family of finite Dyck lattices. Start from all finite paths ordered by padding, adjoin limits. Nothing is excluded; the infinite objects appear as limits and are paths that return to the ground line forever. *Complete Heyting algebra, obtained for free.*	Infinite paths taken as primitive, over a time line enlarged by unbounded stretches. Paths that eventually go quiet are refused: the interesting behaviour must recur forever. *Gains a temporal modality, loses bottom and meet.*
Syntax	$\varphi ::= p \mid \bot \mid \top \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \to \varphi$ *full intuitionistic; $\bot$ and $\wedge$ present, so $\neg\varphi := \varphi \to \bot$ is available. No henceforth: every world is a bounded stretch, so the modality would be inert.*	$\varphi ::= p \mid \top \mid \varphi \vee \varphi \mid \varphi \to \varphi \mid G_i \quad (i \in \omega)$ *positive fragment; $\bot$ and $\wedge$ absent because the carrier is not closed under them. G_i reads* at or after instant i *— meaningful only because rays are worlds.*
Dyck grammar	$E ::= u\,D\,d$, D a finite Dyck word $w = E_0E_1E_2\cdots$ *an ω-product of finite excursions, no side condition. $E_i = ud$ is allowed at every stage, so $v(ud)^\omega$ — and $(ud)^\omega$ itself — are legal words.*	$w = E_0E_1E_2\cdots$ with $E_i \neq ud$ for infinitely many i or $w = E_0\cdots E_m \cdot u^\omega$ *a fairness condition, not expressible by finite lookahead: the syntactic face of the carrier being a proper coinductive subset.*
Functor (co)fixpoints	$D = \mu Y.(1 + Y \times Y)$ *(finite excursions)* $\Delta_{\mathrm{I}} = \nu X.(D \times X)$ *the full final coalgebra — emit a finite excursion, rebound, repeat. Algebra: $\mathcal{H}^{\mathrm{reb}} = \mathcal{O}(\mathrm{Int}_b(\omega))$, a frame.*	$D = \mu Y.(1 + Y \times Y)$ $\Delta_{\mathrm{II}} \subsetneq \nu X.(D \times X + X)$ *the summand $+X$ is the unmatched up-step, whence rays; the carrier is the fair part only. Algebra: $\mathcal{H}^{\mathrm{sup}} = \{D : \mathrm{supp}(D)$ unbounded$\}$, not a sublattice.*
Paths accepted and rejected	**Accepted:** every path returning to the axis infinitely often — including $v(ud)^\omega$ and $(ud)^\omega$ itself, which is the bottom element. **Rejected:** none. **Absent as paths:** the infinite ascents. u^ω and $v \cdot u^\omega$ exist in the algebra only as non-compact limits of chains of finite paths.	**Accepted:** rebound paths whose excursions recur cofinally, and the ray-bearing paths $v \cdot u^\omega$ including $u^\omega = \top$. **Rejected:** every $v(ud)^\omega$ — finite content dressed as an infinite object — and with them the bottom element. **Consequence:** no minimal elements, meets only partial.

The two carriers overlap: the cofinally-nontrivial rebound paths belong to both. What is proper to Model I is the eventually-quiet stratum; what is proper to Model II is the ray stratum as genuine paths. Adding limits costs nothing; removing paths costs $\bot$, $\wedge$ and $\neg$.